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Lec 1

Introduction to Probability Theory

* **Random Experiment:**

e.g. : throwing a coin → may be Head or Tail
H T
" " dice → may be 1, 2, 3, 4, 5, or 6

* **Meaning of probability:**

Throwing a coin : what is the probability of H ?

$P(H) = ?$

⇒ 50% what does that mean ?

Repeating the experiment ∞ times ⇒ 50% of times the outcome of the experiment is H

the concept of relative freq.

the def. or meaning of probability

* How to calculate probability ?

$$P(A) = \frac{N(A)}{N(S)}$$

A : the event we are concerned to know its prob.

S : the sample space

$\downarrow$

the set of ALL possible outcomes of a random experiment

The event is a subset of the sample space satisfying certain cond^{n}

e.g Throwing a dice :

$$P(\underbrace{even}_{\text{event}}) = \frac{N(\{2,4,6\})}{N(\{1,2,3,4,5,6\})} = \frac{3}{6} = \frac{1}{2}$$

$$P(\underbrace{prime}_{\text{event}}) = \frac{N\{2,3,5\}}{N\{1,2,3,4,5,6\}} = \frac{3}{6} = \frac{1}{2}$$

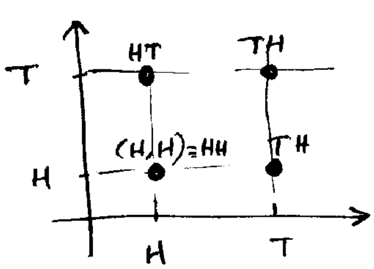
In any random experiment you have to have the skill of determining the <u>Sample space</u>

Some Examples for determing the sample space:

- <u>Ex:</u> Throwing a coin $\Rightarrow$ $S = \{H, T\}$
- <u>Ex:</u> " " dice $\Rightarrow$ $S = \{1, 2, 3, 4, 5, 6\}$

- <u>Ex:</u> Throwing a coin twice

$\Rightarrow$ the problem becomes <u>2D</u>



or

2nd throw \ 1st throw	H	T
H	HH	HT
T	TH	TT

$\Downarrow$

$S = \{HH, HT, TH, TT\}$

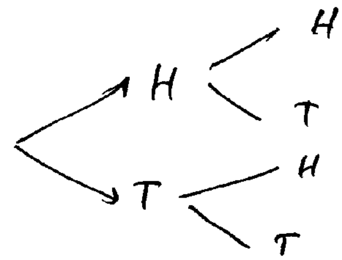
If throw the coin 3 times $\Rightarrow$ 3D

hard to visualize this way

" " " " 4 times $\Rightarrow$ 4D $\rightarrow$ can't be visualized

Alternative way to think about multi-dimensional random experiments $\Rightarrow$ The tree diagram

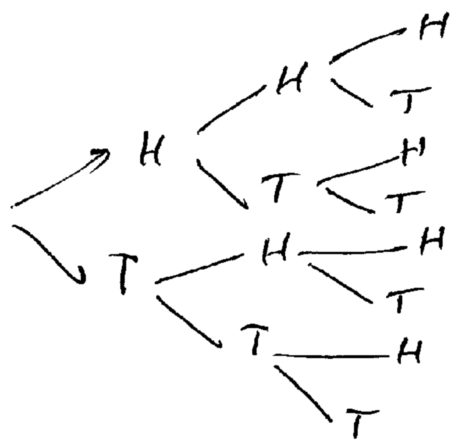
Throw coin twice:



each path is an element in S

$\Rightarrow S = \{HH, HT, TH, TT\}$

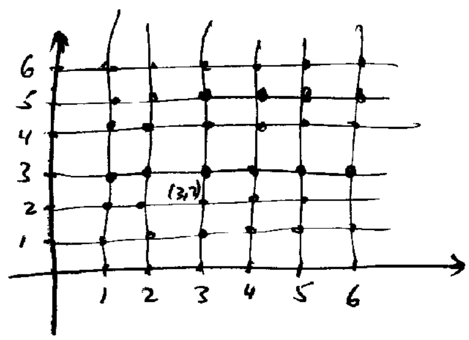
Throw coin 3 times:



$$\Rightarrow S = \{\underbrace{HHH, HHT, \ldots, TTT}_{8 \text{ elements}}\}$$

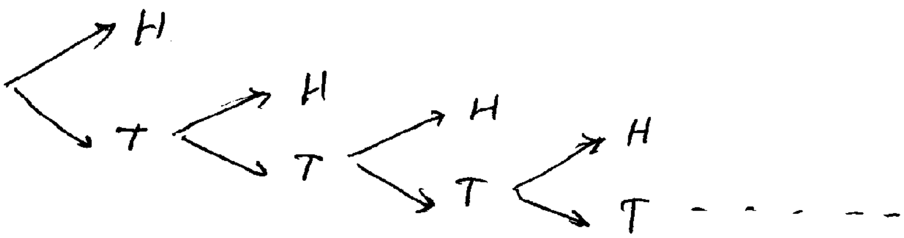
- **Ex: Throw two dices:**

$\Rightarrow$ 2D [tree diagram is messy here]



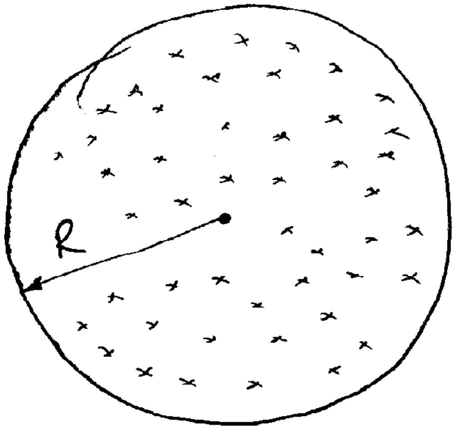
$$S = \underbrace{\{(1,1),(1,2),\ldots\ldots,(6,6)\}}_{36 \text{ elements}}$$

- **Ex: Throw coin until H appears:**



$$S = \underbrace{\{H, TH, TTH, TTTH, \ldots\ldots\}}_{\infty \text{ elements}}$$

Another form for the sample space

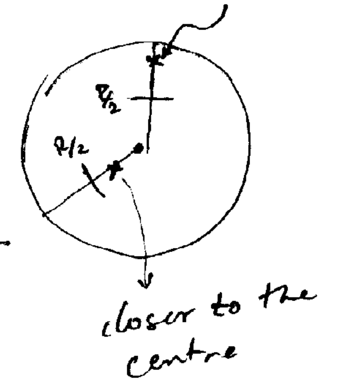


Suppose we want to choose a point randomly in the shown circle

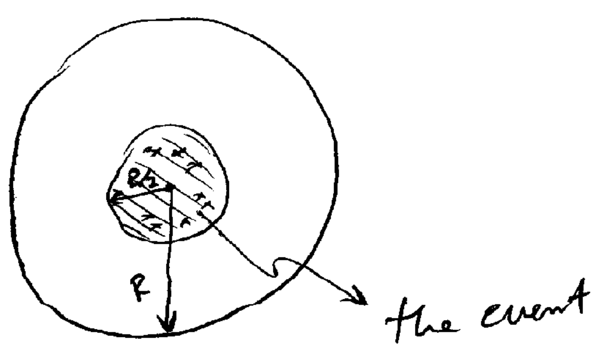
How to find the probability that the chosen point is closer to the centre than it is to the circumference of the circle?

Obviously, the sample space is the set of all points in the circle [infinite n° of elements]

How to represent the event??



The event can be represented by all the points inside the circle with radius $\frac{R}{2}$



Now we want to compute the probability of the event

$$P(\underbrace{\text{chosen point closer to centre}}_{\text{event}}) = \frac{N(\text{event})}{N(S)} = \frac{\infty}{\infty} \leftarrow$$

cannot be calculated this way

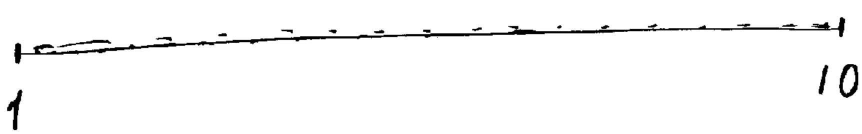
Change the way we calculate prob.

Instead of counting the nº of elements, it is better here to calculate the <u>AREA</u> of the event and the sample space

$$P(\text{event}) = \frac{\text{area}(\text{event})}{\text{area}(S)} = \frac{\pi\left(\frac{R}{2}\right)^2}{\pi(R)^2} = \frac{1}{4}$$

$\underline{\underline{Ex}}$: If we chose a $\underline{\underline{real}}$ nº randomly from the interval $[1, 10]$. Find the prob. that the chosen nº is in the interval $[2, 4]$

$\underline{\underline{Ans}}$: Note that we are not talking about integer nºs but REAL nºs i.e. $1.5, \sqrt{2}, 3.761 \ldots \in [1, 10]$



S = all points in $[1, 10] \Rightarrow \infty$ nº of elements & continuous

Event = all points in $[2, 4] \Rightarrow$ " " " "

If we say $P(\text{event}) = \dfrac{N(\text{event})}{N(S)} = \dfrac{\infty}{\infty} = ?$

$\Rightarrow$ choose another way

$\therefore$ Better measure by LENGTH

$$\therefore P(\text{event}) = \frac{\text{Length } [2,4]}{\text{Length } [1,10]}$$

$$= \frac{4-2}{10-1} = \frac{2}{9} \quad \#$$

As we have seen

S can be discrete & finite nº of elements

e.g. dice $S = \{1,2,3,4,5,6\}$

OR discrete & infinite nº of elements

e.g. throw a coin until H appears

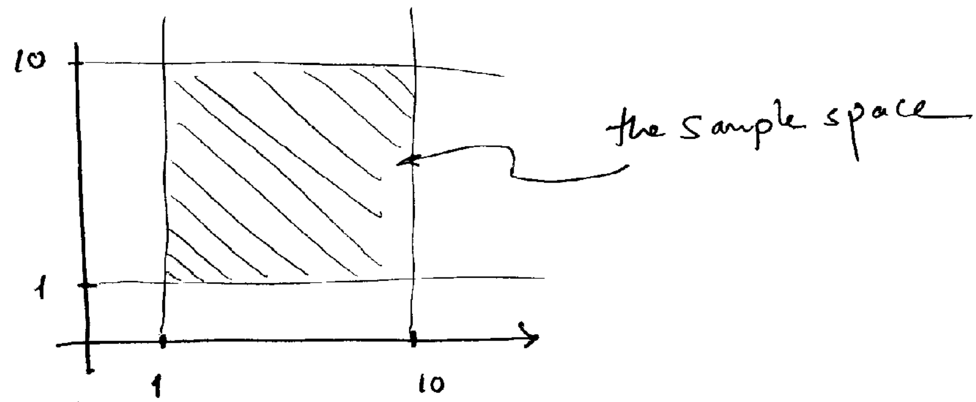
$S = \{H, TH, TTH, TTTH, \dots\}$

OR Continuous & infinite nº of elements

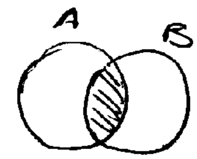
e.g. choose a nº from ~~real~~ real interval $[1, 10]$

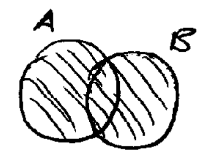
Ex: Suppose we want to choose **two** real n°s from the interval $[1, 10]$. Describe the sample space

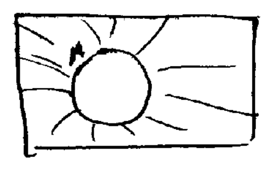
Ans: Just like the problem of throwing 2 coins (OR throwing the coin twice) the problem becomes 2D



Combining events :

$A \cap B \equiv$ in A **And** in B 

$A \cup B \equiv$ in A **OR** in B 

$A^c \equiv$ **NOT** in A 

Ex: When throwing a dice

$$P(\text{prime} \underset{\cap}{\overset{}{\&}} \text{odd}) = P\left(\{2,3,5\} \cap \{1,3,5\}\right)$$
$$= P\left(\{3,5\}\right)$$

$$P(\text{prime} \underset{\cup}{\text{or}} \text{even}) = P\left(\{2,3,5\} \cup \{2,4,6\}\right)$$
$$= P\left(\{2,3,4,5,6\}\right)$$

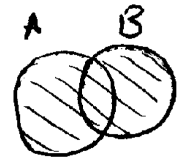
$$P(\text{not prime}) = P(S - A) = P\left(\{1,2,3,4,5,6\} - \{2,3,5\}\right)$$
$$= P\left(\{1,4,6\}\right)$$

Laws of Probability :

i) $P(\phi) = 0$

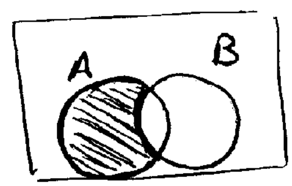
ii) $P(S) = 1$

iii) $P(A^c) = 1 - P(A)$



iv) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

v) $P(A \cap B^c) = P(A) - P(A \cap B)$



in A and not in B

De Morgan Laws : [in Set theory]

$$A^c \cap B^c = (A \cup B)^c$$

$$A^c \cup B^c = (A \cap B)^c$$

How is it used in probability?

$$P(A^c \cap B^c) = P((A \cup B)^c) = 1 - P(A \cup B) =$$

In order to mathematically prove the laws of prob. we must use the Axioms of Probability

Axioms of probability:

I] $0 \leq P(A) \leq 1$

II] $P(S) = 1$

III] If A, B are disjoint events $[A \cap B = \phi]$
then $P(A \cup B) = P(A) + P(B)$

impossible to happen at the same time
e.g outcome of dice even & odd at the same time!

How to use the axioms of probability to prove the laws??

We first describe it as disjoint events then use III

Ex: Prove $P(A^c) = 1 - P(A)$

Ans we know $A \cup A^c = S$

$$\therefore \underbrace{P(A \cup A^c)}_{\text{disjoint}} = P(S)$$

from [III] ; from [II]

$$P(A) + P(A^c) = 1 \Rightarrow P(A^c) = 1 - P(A)$$

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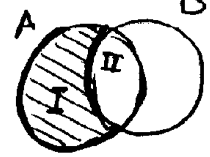
Ex: Prove $P(A\cap B^c) = P(A) - P(A\cap B)$ [*]



Ans:

we know $\underbrace{\underbrace{(A\cap B^c)}_{\mathrm{I}} \cup \underbrace{(A\cap B)}_{\mathrm{II}}}_{\text{disjoint}} = A$

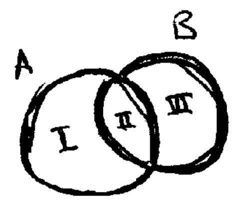
$$P((A\cap B^c)\cup(A\cap B)) = P(A)$$

from III

$$P(A\cap B^c) + P(A\cap B) = P(A)$$

$$\therefore\ P(A\cap B^c) = P(A) - P(A\cap B)$$

Ex: Prove $P(A\cup B) = P(A) + P(B) - P(A\cap B)$



Ans:

$$A\cup B = \underbrace{\underbrace{(A\cap B^c)}_{\mathrm{I}} \cup \underbrace{(A\cap B)}_{\mathrm{II}} \cup \underbrace{(B\cap A^c)}_{\mathrm{III}}}_{\text{disjoint}}$$

$$\therefore P(A\cup B) = \underset{\downarrow\ \text{from}[*]}{P(A\cap B^c)} + P(A\cap B) + \underset{\downarrow\ \text{from}[*]}{P(B\cap A^c)}$$

$$= [P(A) - P(A\cap B)] + \cancel{P(A\cap B)} + [P(B) - \cancel{P(A\cap B)}]$$

$$= P(A) + P(B) - P(A\cap B)$$

$\underline{\underline{Ex:}}$ If $P(B) = 0.5$ & $P(A \cap B^c) = 0.2$

find $P(A \cup B)$, $P(A^c \cup B)$

Also find $P(A)$ in the case of $P(A \cup B^c) = 0.6$

Ans:

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.2$$

$$P(B) = 0.5$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.2 + 0.5 = 0.7 \quad \#$$

De Morgan

$$P(A^c \cup B) = P((A \cap B^c)^c) = 1 - P(A \cap B^c)$$

$$= 1 - 0.2 = 0.8 \quad \#$$

$$P(A \cup B^c) = 1 - P(A^c \cap B) = 1 - [P(B) - P(A \cap B)]$$

$$= 1 - \underbrace{P(B)}_{0.5} + P(A \cap B)$$

$$= 0.5 + P(A \cap B) = 0.6$$

$$\therefore P(A \cap B) = 0.1$$

but since $P(A \cap B^c) = P(A) - \underbrace{P(A \cap B)}_{0.1} = 0.2 \Rightarrow P(A) = 0.3$

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